

A Framework for Evaluating Distributional Diversity in Generative Music Models

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Abstract

Recent generative music systems can synthesize high-quality, stylistically coherent audio, yet it remains unclear to what extent they preserve the *distributional diversity* observed in real musical corpora. We present a systematic framework for evaluating tonal and melodic diversity by extracting musically meaningful representations from real and generated audio and comparing the resulting feature distributions under both in-distribution and out-of-distribution conditions. A range of diversity metrics are applied to quantify concentration, coverage, and distributional drift, revealing consistent mode-seeking tendencies in tonal distributions and stronger deviations under domain shift. These findings motivate diversity-aware evaluation as a complementary lens to perceptual quality and likelihood-based assessments.

1 Introduction

Generative music models have advanced rapidly in audio fidelity and stylistic coherence, but evaluation often emphasizes sample-level quality (e.g., realism, artifact rates) or likelihood-based objectives. Such metrics do not directly capture whether a model reproduces the *variety* present in human music—e.g., diversity across tonal centers, melodic motion, and contour statistics. Moreover, a model can achieve plausible outputs while still collapsing toward a narrow subset of high-probability patterns (“mode seeking”) ??.

This paper proposes a *model-agnostic* evaluation framework that measures diversity as a *corpus-level* property: rather than asking whether a generation matches a specific ground-truth exemplar, we compare the distributions of extracted musical features between real and generated corpora. We focus on two dimensions that admit scalable representations: **key** (global tonal center) and **melody** (main-melody interval and contour statistics).

2 Problem Setup and Definitions

2.1 Diversity as a Distributional Property

Let $\mathcal{D}_{\text{real}}$ be a corpus of real audio clips and $\mathcal{D}_{\text{gen}}$ be a corpus generated by a model under fixed decoding conditions. Let $\phi(\cdot)$ be a representation mapping from audio to musical features. Diversity evaluation compares induced feature distributions $P_{\text{real}}(\phi)$ and $P_{\text{gen}}(\phi)$.

This formulation deliberately distinguishes:

- **Representation / features** (what is extracted): e.g., key labels, interval n -gram histograms, embeddings.
- **Distance / divergence measures** (how distributions are compared): e.g., entropy, KL divergence, overlap mass, EMD.
- **Pairwise similarity** (example-to-example alignment), which can be useful for controllability/fidelity questions but is not equivalent to distributional diversity.

2.2 In-Distribution vs Out-of-Distribution Evaluation

We evaluate diversity under:

- **In-distribution (ID)** reference data: music closer to the model’s training domain.
- **Out-of-distribution (OOD)** reference data: stylistically different music used to test robustness and distributional drift.

(We avoid dataset names in the abstract for readability; they are described in the Evaluation section.)

3 Research Statement and Questions

3.1 Research Statement

We study whether generative music models preserve, compress, or distort the distributional diversity of real music along tonal and melodic dimensions, and how these effects change under domain shift and repeated sampling.

3.2 Research Questions (Reframed)

- **RQ1 (Tonal Diversity Shift):** How do generated key distributions differ from real key distributions in terms of concentration (entropy), coverage, and distributional divergence?
- **RQ2 (Melodic Diversity Shift):** How do generated corpora differ from real corpora in melody-feature distributions (interval motion, contour statistics, and embedding distributions)?
- **RQ3 (Domain Robustness):** How does the diversity gap $P_{\text{real}}(\phi)$ vs. $P_{\text{gen}}(\phi)$ change when the reference data is OOD?
- **RQ4 (Sampling Stability):** How sensitive are diversity estimates to repeated generations under identical prompts and fixed decoding settings?

Note: These questions are explicitly non-binary and do not assume a desired outcome; they quantify shifts and sensitivities.

4 Related Work and Background (Narrative)

Evaluation of generative systems often decomposes into two core choices: (1) selecting a representation that captures relevant content, and (2) selecting a distance/divergence measure to compare distributions. In MIR, tonal analysis and key estimation are supported by chroma-based and tonal centroid features, with toolkits such as *librosa* and *madmom* enabling scalable extraction of global key labels for corpus-level analysis. For melody-related analysis, a wide range of melody extraction and pitch-tracking methods exist, including classic approaches like Melodia Salamon and Gómez 2012, and modern learned representations such as CREPE Kim, Salamon, Li, and Bello 2018 and MERT Li 2023, which provide embedding spaces suitable for similarity and distributional comparison.

To compare musical sequences, prior work has explored symbolic similarity metrics (n -grams, edit distance), geometric measures Aloupis 2004, and alignment costs such as DTW and FastDTW Salvador and Chan 2007. For comparing distributions, Earth Mover’s Distance (EMD) Rubner, Tomasi, and Guibas 2000 is a standard choice when features can be represented as histograms. However, while these tools are well-established for analysis and retrieval, fewer studies use them to evaluate *diversity preservation* in modern generative music systems under controlled experimental settings, especially in ID/OOD comparisons.

5 Method: High-Level Experimental Design (What/Why)

5.1 Why Diversity Might Be a Problem

Generative models can concentrate probability mass on a limited set of patterns, which may yield coherent outputs but reduce coverage of rare events and stylistic variation ???. In music, this can appear as reduced tonal variety and narrower distributions of melodic motion. We therefore measure diversity along dimensions with clear musical meaning and scalable extraction.

5.2 Overview of the Evaluation Pipeline

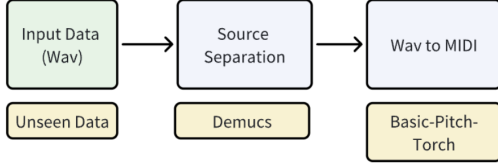
We evaluate two axes: **key diversity** and **melodic diversity**. Figure 1 provides the high-level flow. We intentionally separate *process description* from *implementation tools*; tool choices are listed in an Implementation subsection.

5.3 Experimental Configurations (Clear List)

We use the following recurring configuration schema to avoid ambiguity:

- **System:** Generative model (e.g., MusicGen-melody) + decoding settings.
- **Reference data:** ID corpus vs. OOD corpus.
- **Generation protocol:** number of generations per clip/prompt, fixed randomness, etc.
- **Representation:** key labels; melody features (symbolic + embeddings).
- **Comparison level:** distribution-level (real vs gen); stability (gen vs gen).

Waveform Music to Main Melody Pipeline



Diversity Eval with Generated Text

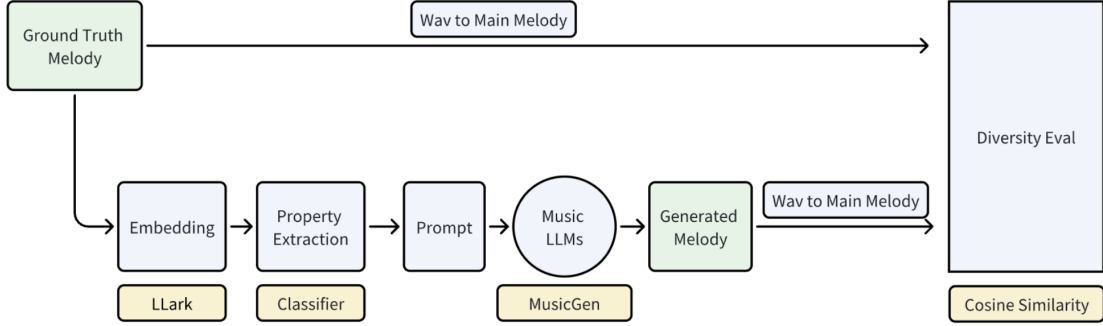


Figure 1: **High-level diversity evaluation pipeline.** We extract tonal and melodic representations from real and generated audio and compare the resulting feature distributions at the corpus level.

6 Key Diversity Pipeline

6.1 Representation (Key Labels)

For each 15 s clip, we estimate a global key using two independent estimators (**librosa** and **madmom**) and retain only clips where the two estimators agree. We refer to such a clip as a **key-labeled clip**: an audio excerpt for which we have a high-confidence global key label derived from estimator agreement (not from manual annotation).

6.2 Distribution-Level Metrics

Let P_{real} and P_{gen} denote empirical distributions over 24 major/minor classes. We compute:

- **Coverage (Total Key Ratio):** proportion of keys observed at least once.
- **Entropy and normalized diversity:** $H(P)$ and $H(P)/\log 24$.
- **KL divergence:** $D_{\text{KL}}(P_{\text{real}} \parallel P_{\text{gen}})$.
- **Overlap mass:** $\sum_i \min(P_{\text{real}}(i), P_{\text{gen}}(i))$.
- **Rare-key activation:** proportion of keys below a probability threshold.

(Here, entropy/coverage characterize concentration; KL/overlap quantify distributional drift.)

7 Melody Diversity Pipeline

7.1 Representation: Waveform-to-Main-Melody

A central challenge for melody diversity is obtaining a scalable representation. We use a two-step waveform-to-main-melody process:

1. **Source separation:** isolate the lead/foreground melodic stem.
2. **Audio-to-symbolic transcription:** transcribe the melodic stem into a symbolic melody representation (notes or pitch contour).

We refer to this as “waveform-to-main-melody” throughout the paper: the output is the symbolic melody representation used for downstream feature extraction.

7.2 Two Complementary Evaluation Modes (Clarifies Confusion)

We distinguish two analyses:

- **(A) Distributional melody diversity (primary):** compare real vs generated corpora in terms of melody-feature distributions (interval motion, contour histograms, embedding distributions). This answers RQ2.
- **(B) Prompt-conditioned melody fidelity / stability (secondary):** when generation is conditioned on a caption derived from an input clip, we measure how consistently the model reproduces related melodic attributes. This is *controllability/fidelity*, not diversity. It answers RQ4-style stability questions (gen vs gen; and sensitivity to prompting).

This separation avoids conflating “high similarity to a described input” with “high diversity”.

7.3 Melody Features vs Distance Measures (Separated)

Features / representations. From each symbolic melody, we compute corpus-level feature distributions such as: interval sequences, interval n -gram histograms, and pitch-class/interval histograms. We also compute learned embeddings (audio or symbolic) to obtain embedding distributions.

Distance measures. We then compare feature distributions using appropriate measures:

- **Distribution distances** (distribution-wise): EMD for histograms, KL (where appropriate), overlap mass, etc.
- **Pairwise distances** (pairwise): edit distance and DTW are used only in Mode (B) or when explicitly aggregating pairwise costs into a distributional statistic (e.g., average within a generation set).

Note duration for DTW. When DTW is used, we apply it to time-stamped pitch contours or note sequences that include durations; alignment is computed on sequences where each element is expanded or weighted by duration (details can be specified once the exact implementation is fixed).

8 Evaluation: Data, Systems, and Protocol

8.1 Systems

Our current experiments evaluate MusicGen-melody as the generative system. The framework is model-agnostic and can be applied to other systems producing audio outputs.

8.2 Reference Data (ID vs OOD)

We use:

- **In-distribution reference:** Shutterstock Solo Instrumental.
- **Out-of-distribution reference:** BJ Opera.

8.3 Generation Procedure (Clear and Consistent)

Key diversity. Each key-labeled clip is regenerated three times under fixed decoding settings to assess sampling variability. We compute P_{gen} over all generations and also report per-generation ranges where relevant.

Melody diversity (distributional). We extract melody features from real corpora and from generated corpora, then compare corpus-level distributions (Mode A).

Prompt-conditioned analysis (optional / secondary). When using structured captions (e.g., from Qwen3-Omni) to prompt generation, we treat the resulting comparisons as controllability/fidelity diagnostics (Mode B), not as the primary diversity metric.

9 Results and Discussion (Phase I: Key Diversity)

9.1 Key Diversity (Phase I)

In-distribution findings. Shutterstock exhibits broad key coverage (ratio 0.96) with high entropy ($H = 4.12$). MusicGen generations maintain partial coverage (0.83–0.96) but show reduced entropy (3.85–3.98) and diversity (0.84–0.87). KL divergence remains modest (0.25–0.82), indicating partial preservation of the ID distribution while collapsing toward dominant tonal centers.

Out-of-distribution findings. BJ Opera has higher entropy ($H = 4.20$). MusicGen outputs shift distribution mass toward Western tonal centers, reducing overlap (0.56) and increasing KL divergence (1.75). This drift suggests strong training-domain priors and weaker cross-domain robustness.

Presentation note. OOD results will be visualized as key histograms to improve interpretability (see planned Figure placeholders below).

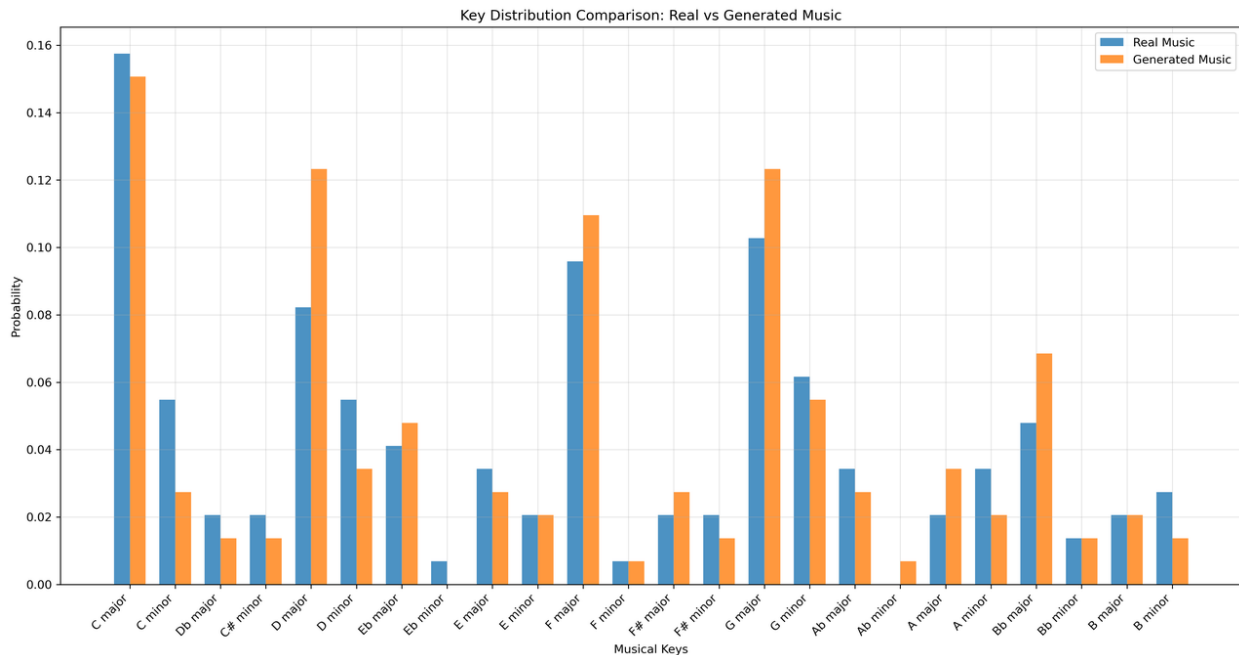


Figure 2: Key histogram comparison for in-distribution data: real vs generated.

10 Implementation Notes (Moved out of Results)

10.1 Main-Melody Transcription Tooling

We benchmarked several main-melody transcription tools to select a robust default backend for large-scale processing. Pop2Piano and Basic-Pitch-Torch performed reliably in practice, with Pop2Piano producing simplified piano-style reductions and Basic-Pitch-Torch preserving pitch contours more faithfully on diverse audio. MR-MT3 variants were difficult to deploy at scale due to dependency/weight conversion issues. Based on these observations, we adopt Basic-Pitch-Torch as the default transcription backend and use Pop2Piano as a qualitative reference when needed.

(These details are included here to justify the representation pipeline, rather than as a result about model diversity.)

11 Planned Phase II (Melody Diversity)

With the transcription backbone fixed, our next step is to quantify melody diversity using the distributional design in Section 7. We will report both: (i) real vs generated distribution distances for interval and embedding features, and (ii) sampling stability metrics across repeated generations with the same conditioning.

12 Novelty of the Proposed Work

- **Model-agnostic diversity evaluation:** a framework that measures diversity as a corpus-level property using musically meaningful representations and distributional comparisons.
- **Cross-domain analysis:** explicit ID/OOD evaluation revealing distributional drift patterns

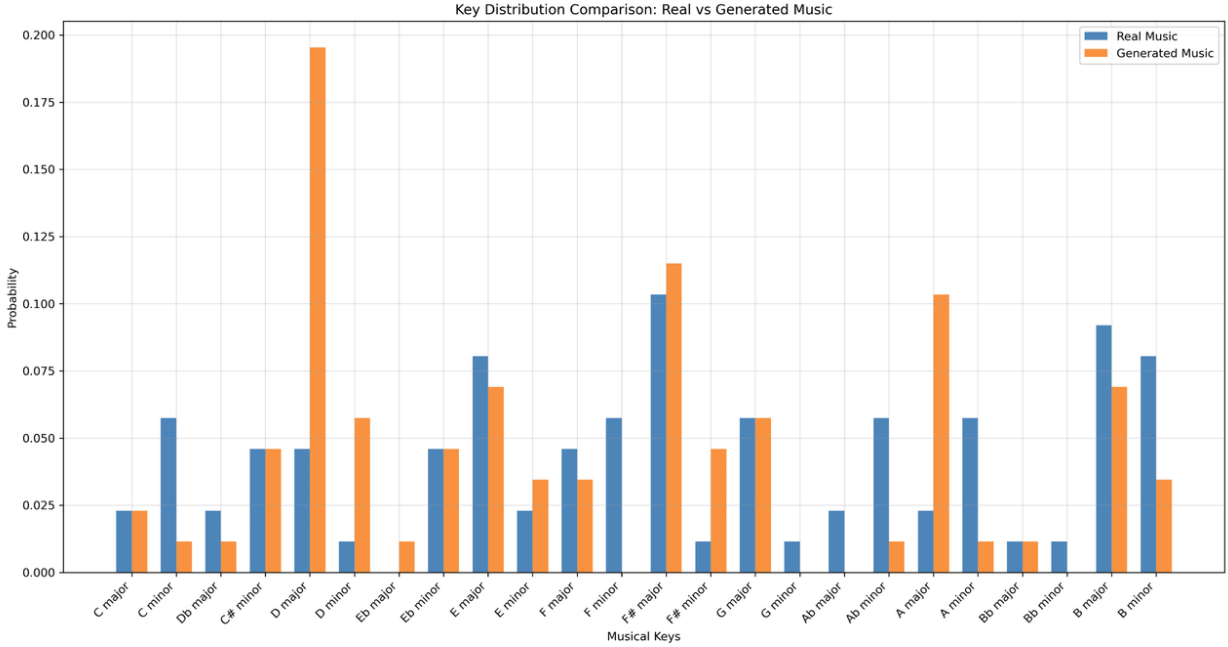


Figure 3: Key histogram comparison for out-of-distribution data: real vs generated.

under domain shift.

- **Unified pipeline:** linking waveform processing, melody representation extraction, caption-based prompting (optional), and distributional diversity scoring.

13 Deliverables

- A complete evaluation pipeline covering key diversity and distribution-level melodic diversity, integrating source separation, melody transcription, caption generation, music synthesis, and diversity scoring.
- A fully processed dataset of 337 key-labeled Shutterstock clips with three corresponding MusicGen generations per clip for key-variance analysis.
- A large-scale captioned corpus for melody analysis: 500 audio clips source-separated and captioned using Qwen3-Omni; 300 captions used to generate MusicGen outputs, which were again source-separated and transcribed for downstream evaluation.
- A scalable Qwen3-Omni captioning system (via Apptainer + VLLM) capable of captioning 30-second audio clips in under 3 seconds using 4×H100/H200 GPUs.

14 Conclusion

We presented a model-agnostic framework for evaluating distributional diversity in generative music systems, focusing on tonal and melodic dimensions. Phase I results indicate tonal mode seeking

in-distribution and stronger distributional drift out-of-distribution. Ongoing Phase II work will quantify melodic diversity using corpus-level feature distributions and embedding distances, with controllability/stability analyses reported separately to avoid conflating pairwise fidelity with diversity.

References

- Rubner, Yossi, Carlo Tomasi, and Leonidas J. Guibas (2000). “The Earth Mover’s Distance as a Metric for Image Retrieval”. In: *International Journal of Computer Vision* 40.2, pp. 99–121.
- Aloupis, Greg (2004). “Algorithms for Computing Geometric Measures of Melodic Similarity”. In: *Proceedings of the International Symposium on Algorithms and Computation*. Springer, pp. 308–319.
- Salvador, Stan and Philip Chan (2007). “FastDTW: Toward Accurate Dynamic Time Warping in Linear Time and Space”. In: *Intelligent Data Analysis* 11.5, pp. 561–580.
- Salamon, Justin and Emilia Gómez (2012). “Melody Extraction from Polyphonic Music Signals Using Pitch Contour Characteristics”. In: *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 20.6, pp. 1759–1770.
- Kim, Jong Wook, Justin Salamon, Peter Li, and Juan Pablo Bello (2018). “CREPE: A Convolutional Representation for Pitch Estimation”. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 161–165.
- Li, Yi (2023). “MERT: Acoustic Music Understanding Model with Large-Scale Self-Supervised Training”. In: *arXiv preprint arXiv:2306.00107*.

Appendix A: Key Diversity Metrics

Table 1: Key diversity metrics for the in-distribution and out-of-distribution datasets and their MusicGen generations. “Gen1–Gen3” denote three independent generations for the ID inputs.

Metric	Uniform	In-Distribution				Out-of-Distribution	
		Ori	Gen1	Gen2	Gen3	Ori	Gen
Total key ratio	1.000	0.958	0.833	0.958	0.875	0.875	0.917
Entropy	4.585	4.116	3.852	3.980	3.875	4.195	4.209
Diversity ($H/\log 24$)	1.000	0.898	0.840	0.868	0.845	0.915	0.918
KL to uniform	0.000	1.512	0.250	0.817	1.697	1.748	3.716
Overlap w. uniform	1.000	0.842	0.870	0.836	0.705	0.558	0.532
Rare-key rate	1.000	0.696	0.650	0.739	0.762	0.619	0.682

Appendix B: Notes on Pairwise vs Distributional Metrics (Clarification)

Some measures in the literature are naturally pairwise (e.g., edit distance, DTW), while others compare distributions (e.g., EMD on histograms). In this work, pairwise distances are only used when (i) explicitly answering controllability or stability questions, or (ii) aggregated into a corpus-level statistic (e.g., average within-generation distance). Distribution-level comparisons remain the primary definition of diversity throughout the paper.